

Automated Earnings Quality Agent — Task-by-Task Build Plan

PHASE 1 — PROJECT FOUNDATION

- **Task 1 — Create the project structure**

- Create the main project folder.
- Create separate folders for:
 - data
 - calculations
 - analysis
 - excel_reports
 - tests
- Create the basic Python files.
- Do not build any finance calculations yet.

- **Task 2 — Create the standard financial data structure**

- Define how every financial value will be stored.
- Required fields:
 - Company
 - Ticker
 - Fiscal Year
 - Metric
 - Value
 - Units
 - Financial Statement
 - Source
 - Source Date
- Test with fake/sample data first.

- **Task 3 — Create a manual input test dataset**

- Choose one public company.
- Manually enter 3–5 years of:
 - Revenue

- Net Income
- OCF
- AR
- Inventory
- AP
- D&A
- CapEx
- SBC
- Shares Outstanding
- Tax Expense
- DTA
- DTL

- This dataset becomes our verified test case.

PHASE 2 — FINANCIAL CALCULATION ENGINE

Objective 1 — Accrual vs Cash

- **Task 4 — Revenue growth calculation**

- Calculate YoY Revenue growth.
- Show formula.
- Show current-year value.
- Show prior-year value.
- Show calculated result.

- **Task 5 — Accounts Receivable growth**

- Calculate YoY AR growth.
- Compare AR growth to Revenue growth.
- Do not create warnings yet.

- **Task 6 — Net Income growth**

- Calculate YoY NI growth.

- **Task 7 — Operating Cash Flow growth**

- Calculate YoY OCF growth.
 - **Task 8 — Accrual vs cash comparison table**
 - Create a table showing:
 - Revenue growth
 - AR growth
 - NI growth
 - OCF growth
 - One row per fiscal year.
 - **Task 9 — First accrual red-flag rules**
 - Flag when AR grows materially faster than Revenue.
 - Flag when NI rises while OCF falls.
 - Initially use simple rules that we approve manually.
 - Show the exact rule that triggered each flag.
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PHASE 3 — D&A, CAPEX AND WORKING CAPITAL

- **Task 10 — D&A calculations**
 - Calculate:
 - D&A growth
 - D&A / Revenue
 - Build a D&A trend table.
- **Task 11 — CapEx calculations**
 - Calculate:
 - CapEx growth
 - CapEx / Revenue
 - CapEx / D&A
- **Task 12 — Free Cash Flow**
 - Calculate:
 - $FCF = OCF - CapEx$
 - Show all input values and formula.

- **Task 13 — Working capital movements**

- Calculate yearly changes in:
 - AR
 - Inventory
 - AP
- Classify:
 - AR increase = cash use
 - Inventory increase = cash use
 - AP increase = cash benefit

- **Task 14 — Working capital analysis table**

- Show:
 - AR change
 - Inventory change
 - AP change
 - Net working-capital cash effect

- **Task 15 — CapEx analyst rule**

- Do NOT classify high CapEx automatically as bad.
- If CapEx is unusually high relative to D&A, output:
 - “Needs investigation”
- Later AI can investigate whether it relates to:
 - maintenance
 - expansion
 - factories
 - technology
 - acquisitions

- **Task 16 — Inventory warning rules**

- Compare Inventory growth to Revenue growth.
- Flag unusual divergence.
- Do not automatically conclude demand is weak.

- **Task 17 — AP warning rules**

- Compare AP growth with Revenue, OCF and liquidity data where available.
 - Separate:
 - normal supplier financing
 - possible payment pressure
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PHASE 4 — DEFERRED TAXES

- **Task 18 — Deferred tax data structure**

- Add:
 - DTA
 - DTL
 - Current Tax Expense
 - Deferred Tax Expense
 - Tax-loss carryforwards where available

- **Task 19 — DTA analysis**

- Show DTA trend.
- Compare DTA to profitability.
- Flag large DTA when the company has weak or negative taxable/profit trends.
- Do not automatically assume the full DTA is valuable.

- **Task 20 — DTL analysis**

- Show DTL trend.
 - Identify whether DTL is rising or falling.
 - Keep interpretation conservative until source details are available.
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PHASE 5 — STOCK-BASED COMPENSATION

- **Task 21 — SBC calculations**

- Calculate:
 - SBC growth

- SBC / Revenue
- SBC / OCF
- SBC / Net Income
- **Task 22 — Share dilution calculations**
 - Calculate YoY Shares Outstanding growth.
- **Task 23 — SBC vs dilution analysis**
 - Compare:
 - SBC growth
 - Shares Outstanding growth
 - Flag significant dilution.
- **Task 24 — Buyback adjustment**
 - Add Share Repurchases when available.
 - Check whether buybacks offset SBC dilution.
 - Keep cash spent on buybacks visible.

PHASE 6 — NORMALIZED EARNINGS

- **Task 25 — One-off item data structure**
 - Create categories for:
 - Asset-sale gains
 - Restructuring
 - Legal settlements
 - Impairments
 - Acquisition-related charges
 - Unusual tax gains/losses
 - Other unusual items
- **Task 26 — Manual one-off classification**
 - Before using AI, allow us to manually mark:
 - recurring
 - non-recurring

- uncertain

- **Task 27 — Normalized Net Income calculation**

- Start with Reported NI.
- Subtract genuine one-time gains.
- Add back genuine one-time expenses.
- Calculate Normalized NI.

- **Task 28 — Reported vs Normalized Earnings**

- Show both numbers side by side.
- Calculate the percentage difference.

- **Task 29 — Repeated one-off detection**

- Check whether the same type of “one-time” item appears repeatedly across years.
- If recurring, flag it for review instead of automatically excluding it.

PHASE 7 — RED-FLAG ENGINE

- **Task 30 — Create the red-flag table**

- Every flag must show:
 - Metric
 - Raw data
 - Calculation
 - Rule
 - Result
 - Severity
 - Explanation

- **Task 31 — Accrual red flags**

- AR growth vs Revenue growth
- NI growth vs OCF growth

- **Task 32 — Working-capital red flags**

- Inventory growth vs Revenue
- AP growth vs business growth

- Cash trapped in working capital
 - **Task 33 — SBC red flags**
 - Large SBC
 - Rising share count
 - Buybacks required to offset dilution
 - **Task 34 — Normalization red flags**
 - Repeated “one-time” items
 - Large difference between reported and normalized earnings
 - **Task 35 — Tax red flags**
 - Large DTA with weak profitability
 - Unusual deferred-tax movements
 - **Task 36 — Severity system**
 - Initially use only:
 - Low Risk
 - Needs Investigation
 - Material Concern
 - Do not create a 0–100 score yet.
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PHASE 8 — TABLES AND CHARTS

- **Task 37 — Master analysis table**
 - Create one table containing the important values and calculations across all years.
- **Task 38 — NI vs OCF chart**
- **Task 39 — Revenue vs AR growth chart**
- **Task 40 — Revenue vs Inventory growth chart**
- **Task 41 — D&A vs CapEx chart**
- **Task 42 — OCF vs FCF chart**
- **Task 43 — SBC trend chart**
- **Task 44 — Shares Outstanding trend chart**
- **Task 45 — Reported NI vs Normalized NI chart**

Only keep charts that actually help the analysis.

PHASE 9 — EXCEL AUTOMATION

- **Task 46 — Create basic Excel export**
 - Export analysis to .xlsx.
 - **Task 47 — Raw Financial Data sheet**
 - **Task 48 — Growth Calculations sheet**
 - **Task 49 — Accrual & Cash Analysis sheet**
 - **Task 50 — Working Capital sheet**
 - **Task 51 — D&A and CapEx sheet**
 - **Task 52 — Deferred Taxes sheet**
 - **Task 53 — SBC & Dilution sheet**
 - **Task 54 — Normalized Earnings sheet**
 - **Task 55 — Red Flags sheet**
 - **Task 56 — Dashboard sheet**
 - **Task 57 — Analyst Conclusion sheet**
 - **Task 58 — Excel formulas**
 - Where practical, keep visible Excel formulas instead of exporting only hardcoded calculated values.
 - **Task 59 — Excel formatting**
 - Professional finance formatting.
 - Clear headers.
 - Consistent units.
 - Number formatting.
 - Conditional formatting for flags.
 - **Task 60 — Pivot tables**
 - Add only when the data structure makes them useful.
 - Do not force pivots into the first version.
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PHASE 10 — STREAMLIT APPLICATION

- **Task 61 — Basic Streamlit page**
 - Company input
 - Number of years
 - Analyze button
 - **Task 62 — Financial Data tab**
 - **Task 63 — Cash Conversion tab**
 - **Task 64 — Working Capital tab**
 - **Task 65 — Taxes tab**
 - **Task 66 — SBC tab**
 - **Task 67 — Normalized Earnings tab**
 - **Task 68 — Red Flags tab**
 - **Task 69 — Download Excel button**
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PHASE 11 — AUTOMATIC DATA EXTRACTION

- **Task 70 — Choose reliable financial source**
 - Prefer:
 - company Investor Relations
 - official regulatory filings
 - Do not rely on random finance websites for core numbers.
- **Task 71 — Ticker/company identification**
- **Task 72 — Automatically retrieve annual financial statements**
- **Task 73 — Extract Revenue**
- **Task 74 — Extract Net Income**
- **Task 75 — Extract OCF**
- **Task 76 — Extract AR, Inventory and AP**
- **Task 77 — Extract D&A and CapEx**
- **Task 78 — Extract SBC**

- **Task 79 — Extract Shares Outstanding**
 - **Task 80 — Extract deferred-tax information**
 - **Task 81 — Extract potential one-offs**
 - **Task 82 — Store source for every extracted value**
 - Every number must have:
 - filing/source
 - fiscal year
 - units
 - extraction location if possible
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PHASE 12 — DATA VALIDATION

- **Task 83 — Missing-data validation**
 - **Task 84 — Unit validation**
 - **Task 85 — Duplicate-value validation**
 - **Task 86 — Fiscal-year validation**
 - **Task 87 — Label mapping**
 - Example:
 - Accounts Receivable
 - Receivables
 - Trade Receivables
 - Map them into a standard metric where appropriate.
 - **Task 88 — Manual correction screen**
 - Allow the user to correct an extracted value before analysis.
 - **Task 89 — Stop analysis if critical data fails validation**
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PHASE 13 — AI INTERPRETATION

- **Task 90 — Create structured AI input**
 - AI receives calculations and flags.

- AI does not calculate core numbers itself.
- **Task 91 — Accrual interpretation**
- **Task 92 — Working-capital interpretation**
- **Task 93 — CapEx interpretation**
- **Task 94 — Deferred-tax interpretation**
- **Task 95 — SBC interpretation**
- **Task 96 — Normalized earnings interpretation**
- **Task 97 — Final earnings-quality conclusion**
 - Strong
 - Generally Healthy
 - Needs Investigation
 - Weak
 - Explanation required.
- **Task 98 — Prevent unsupported conclusions**
 - Never automatically state:
 - fraud
 - manipulation
 - insolvency
 - Use language such as:
 - possible
 - may indicate
 - requires investigation

PHASE 14 — ACCURACY TESTING

- **Task 99 — Create manual verified answer sheet**
- **Task 100 — Test raw data accuracy**
- **Task 101 — Test every calculation**
- **Task 102 — Test every red-flag rule**
- **Task 103 — Test normalized earnings**

- **Task 104 — Test Excel output**
 - **Task 105 — Test AI explanations**
 - **Task 106 — Test difficult scenarios**
 - High inventory before product launch
 - High CapEx for expansion
 - High AP from supplier bargaining power
 - High SBC with offsetting buybacks
 - Large DTA with strong future earnings
 - Repeated “one-time” charges
 - **Task 107 — Test second company**
 - **Task 108 — Test third company from another non-financial industry**
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PHASE 15 — FINAL AGENT

- **Task 109 — Connect all modules to one controller**
 - User enters company.
 - Agent executes modules in correct order.
 - **Task 110 — Add progress tracking**
 - Data extraction
 - Validation
 - Calculations
 - Red flags
 - AI interpretation
 - Excel generation
 - **Task 111 — Add failure handling**
 - If something fails, tell the user exactly which stage failed.
 - **Task 112 — Final downloadable analysis**
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DEVELOPMENT RULE

For every task:

1. Define exactly what the task should do.
2. Give Codex only that task.
3. Codex writes the code.
4. Run the code.
5. Test the output manually.
6. Compare expected vs actual.
7. Fix errors.
8. Approve the task.
9. Commit/save the working version.
10. Only then move to the next task.

Never give Codex several untested modules at once.

Later extensions:

- ❓ **Quarterly mode** → 10-Q, Q1/Q2/Q3, standalone-quarter vs YTD handling
- ❓ **Non-U.S. companies** → official regulator/IR sources outside SEC
- ❓ **Optional filing-link input** → user can paste a specific annual/quarterly filing
- ❓ **Broader company lookup** → resolve companies outside the small local validation set
- ❓ **Dynamic app controls** → company/ticker, Annual/Quarterly, number of periods, Analyze
- ❓ **Route the same analysis engines** to whichever validated dataset the user selected